

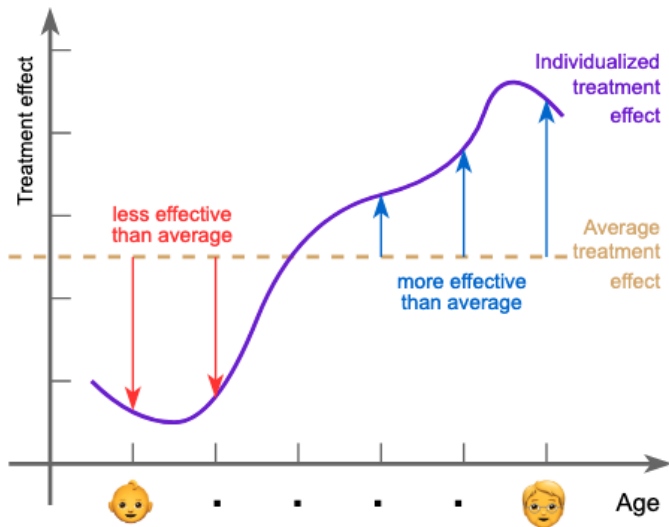
Heterogeneous Treatment Effects and Causal Forest

Zhiyuan Chen

Empirical Methods in Economics and Business
Remin Business School

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Why Heterogeneous Treatment Effects?



Feuerriegel, Stefan, et al. “Causal machine learning for predicting treatment outcomes.”

Nature Medicine 30.4 (2024): 958-968.

Outline

- 1 Motivation
- 2 Causal Inference Basics
- 3 Causal Trees
- 4 Applications and Extensions

Why Causal ML?

- Traditional economics models: homogeneous effects, parametric.
- Many applications require individual-level (or subgroup) treatment effects: *personalized medicine, targeted marketing, policy evaluation*.
- High-dimensional covariates and complex interactions challenge classical nonparametric methods.
- Goal: flexible, data-driven estimation with valid inference.

Potential Outcomes Framework

- For unit i : covariates X_i , treatment $D_i \in \{0, 1\}$, outcomes Y_i^0, Y_i^1 .
- **Treatment effect** at x :

$$\tau(x) = E[Y_i^1 - Y_i^0 \mid X = x]$$

- Observed: $Y_i = Y_i^{D_i}$; missing counterfactual Y_i^0 for treated units.
- **Unconfoundedness/Conditional Independence:**

$$\{Y_i^0, Y_i^1\} \perp D_i \mid X_i.$$

- **Overlap:** $\varepsilon < e(x) = P(D_i = 1 \mid X_i = x) < 1 - \varepsilon$, where $e(x)$ is propensity score of receiving treatment.
- The individual treatment effect is

$$E \left[Y_i \left(\frac{D_i}{e(x)} - \frac{1 - D_i}{1 - e(x)} \right) \right] = \tau(x)$$

Regression Trees as Nearest-Neighbor Matching Method

Given a point x :

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 - ✓ **Adaptive Partitioning:** (1) Make fine splits of the *strong features* capturing targeted signal, (2) make few splits for *weak features*.

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 **Example:** In online ad analysis, suppose we have ten user features like browsing time, device, region, age, and so on but only browsing time exhibits *a sharp jump in conversion once users exceed five minutes*. A regression tree will automatically carve out several *narrow leaves around that 5-minute threshold* to capture the rapid change in ad effectiveness, while keeping leaves broad for other features where conversion varies slowly.

Regression Trees Adjusted for Treatment Effects

Wager and Athey (2017, JASA)

We observe independent samples (X_i, Y_i) and want to build a CART (Classification and Regression Tree):

- 1 **Recursive Partitioning:** recursively splitting X -space into L leaves, each of which containing a few training samples.
- 2 **Evaluating Prediction**(as in CART): given a text point x , identify the leaf $L(x)$ containing x and predict

$$\hat{\mu}(x) = \frac{1}{|\{i : X_i \in L(x)\}|} \sum_{i: X_i \in L(x)} Y_i$$

- 3 **Estimating Treatment Effects:**

$$\hat{\tau}(x) = \underbrace{\frac{1}{|\{i : D_i = 1, X_i \in L(x)\}|} \sum_{i: D_i=1, X_i \in L(x)} Y_i}_{\text{treated}} - \underbrace{\frac{1}{|\{i : D_i = 0, X_i \in L(x)\}|} \sum_{i: D_i=0, X_i \in L(x)} Y_i}_{\text{untreated}} \quad (1)$$

From Causal Trees to Causal Forest

- Use the *causal forest algorithm* to generate an ensemble of B such trees to obtain $\{\hat{\tau}_b(x)\}_{b=1}^B$ and average them:

$$\hat{\tau}(x) = \frac{1}{B} \sum_{b=1}^B \hat{\tau}_b(x)$$

- Individual causal trees are built using random subsamples of s training examples, $s \ll n$

Asymptotic Inference with Causal Forests

- Under Lipschitz conditions and overlap:

$$\frac{\hat{\tau}(x) - \tau(x)}{\sqrt{\text{Var}[\hat{\tau}(x)]}} \rightarrow_d \mathcal{N}(0, 1).$$

- ✓ **Lipschitz condition:** $E[Y^1|X=x]$ and $E[Y^0|X=x]$ are both *Lipschitz continuous*.

- Variance estimated by **Infinitesimal Jackknife**:

$$\hat{V}_{IJ}(x) = \underbrace{\frac{n-1}{n} \left(\frac{n}{n-s} \right)^2}_{\text{finite-sample correction}} \sum_i \widehat{\text{Cov}}^2[\hat{\tau}_b(x), 1_{\{i \in b\}}].$$

- ✓ The covariance is taken over the set of all trees
- ✓ Consistency: $\hat{V}_{IJ}(x) / \text{Var}[\hat{\tau}(x)] \rightarrow_p 1$.

- Confidence intervals: $\hat{\tau}(x) \pm z_{\alpha/2} \sqrt{\hat{V}_{IJ}(x)}$.

Classical Jackknife (nonparametric)

Basic Idea: Measures how sensitive the statistic is to each observation.

Brief History

- **Goal:** Estimate variance (or bias) of a statistic without analytical formulas.
- Jackknife resamples by **leaving one observation out at a time**.
- For sample of size n , define:

$$\hat{\theta}_{(-i)} = s(X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n)$$

- Jackknife variance estimate:

$$\widehat{\text{Var}}_{\text{jack}}(\hat{\theta}) = \frac{n-1}{n} \sum_{i=1}^n (\hat{\theta}_{(-i)} - \bar{\theta}_{\text{jack}})^2$$

$$\bar{\theta}_{\text{jack}} = \frac{1}{n} \sum_{i=1}^n \hat{\theta}_{(-i)}$$

Bootstrap

- Resamples **with replacement** from the original dataset.
- For each bootstrap sample, compute the statistic:

$$\hat{\theta}^{(b)}, \quad b = 1, \dots, B$$

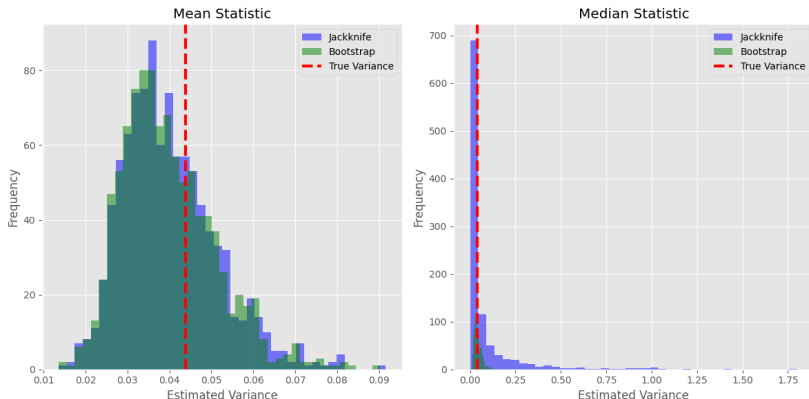
- Bootstrap variance estimate:

$$\widehat{\text{Var}}_{\text{boot}}(\hat{\theta}) = \frac{1}{B-1} \sum_{b=1}^B (\hat{\theta}^{(b)} - \bar{\theta}_{\text{boot}})^2$$

$$\bar{\theta}_{\text{boot}} = \frac{1}{B} \sum_{b=1}^B \hat{\theta}^{(b)}$$

Comparison of Jackknife and Bootstrap

Example: Generates $R=1000$ samples from an exponential distribution. For each sample, estimates variance of: the **mean (smooth)** and **median (non-smooth)**



Note: **Mean:** true = 0.0438, jackknife = 0.0401, bootstrap = 0.0397.

Median: true = 0.0402, **jackknife = 0.0767**, bootstrap = 0.0443.

Jackknife vs. Bootstrap

- Jackknife can be seen as a **linear approximation** to the bootstrap.
- For the sample mean, both give the same variance estimate.
- Bootstrap is more robust for:
 - ▶ Distributional inference (e.g., confidence intervals)
 - ▶ Nonlinear or non-smooth statistics
- Jackknife is interpretable and efficient for basic statistics.

Feature	Jackknife	Bootstrap
Resampling method	Leave-one-out	With replacement
Number of resamples	n	B (large)
Accuracy	First-order approx.	Higher-order approx.
Good for	Smooth statistics	Smooth + non-smooth statistics
Computation	Fast, simple	Slower, flexible

When to Use Which?

- Use **jackknife** when:
 - ▶ The statistic is smooth and differentiable
 - ▶ You need fast, interpretable variance estimates
- Use **bootstrap** when:
 - ▶ You need distributional inference (CIs, p-values)
 - ▶ The statistic is non-smooth or highly nonlinear
- Both methods are foundations for modern **influence-function-based techniques**.

Infinitesimal Jackknife (IJ)

Infinitesimal Jackknife (IJ):

- The IJ is a linear approximation of how the prediction changes as the weight of each observation is perturbed by a tiny amount.
- Instead of recomputing the prediction multiple times, the IJ approximates how the prediction would change if each observations frequency in the dataset increased infinitesimally.
- Think of it as estimating a **sensitivity or influence function**: *how sensitive is the forest prediction to small changes in the training data?*

This method is computationally efficient and can be used with random forests without retraining.

Honest Trees and Forests

A tree is honest if, for each training sample, it **ONLY** uses the response Y_i to estimate the within-leaf treatment effect τ or to decide where to place the splits, *but not both*.

Why Honesty Matters:

- Using the same data for both splitting and estimation can lead to **overfitting** and **biased** treatment effect estimates.
- Honest trees avoid this by mimicking a randomized experiment within each leaf.

Typical Implementation:

- Randomly split the training data:
 - ▶ One part for tree construction (splitting),
 - ▶ One part for treatment effect estimation.

Benefits:

- Reduces estimation bias,
- Enables valid confidence intervals for $\tau(x)$,
- Supports reliable inference of heterogeneous treatment effects.

Why Honest Trees? An Intuition from Causal Inference

Core Problem: Estimating causal effects requires avoiding bias from using **outcome noise** Y_i to both:

- Determine where to split (form subgroups), and
- Estimate treatment effects $\hat{\tau}$ within those subgroups.

Without honesty:

- Tree splits may overfit to *idiosyncratic noise* in Y_i ,
- Outcome-based subgrouping introduces a form of **endogeneity** treatment estimates are biased due to feedback from noise,
- Analogous to **sample selection bias**: group definitions depend on observed (noisy) outcomes.

Algorithm 1: DOUBLE-SAMPLE TREES

Goal: Construct an honest tree by using separate data for splitting and estimation:

- n training samples: (X_i, Y_i) for regression, or (X_i, Y_i, W_i) for causal inference.
- Minimum leaf size: k .

Algorithm Steps:

- 1 **Subsample:** Randomly select a subsample of size $s \in \{1, \dots, n\}$, then split it into:
 - ▶ $\mathcal{E}$: Estimation sample (size $\approx s/2$),
 - ▶ $\mathcal{S}$: Splitting sample (size $\approx s/2$).
- 2 **Grow Tree:** Use only $\mathcal{S}$ to choose tree splits:
 - ▶ Use X_i and W_i from $\mathcal{E}$, but *not* Y_i ,
 - ▶ Ensure each leaf contains at least k units from $\mathcal{E}$.
- 3 **Estimate Responses:** Use only $\mathcal{E}$ to compute:
 - ▶ Leafwise means $\hat{\mu}(x)$ for regression,
 - ▶ Treatment effects $\hat{\tau}(x)$ for causal trees.

Note: For causal trees, each leaf must contain at least k observations *per treatment group* from $\mathcal{E}$.

Algorithm 2: PROPENSITY TREES

Goal: Use variation in treatment assignment W_i to guide splits and estimate conditional treatment effects $\tau(x)$:

- n training examples: (X_i, Y_i, W_i) ,
- Minimum leaf size k .

Algorithm Steps:

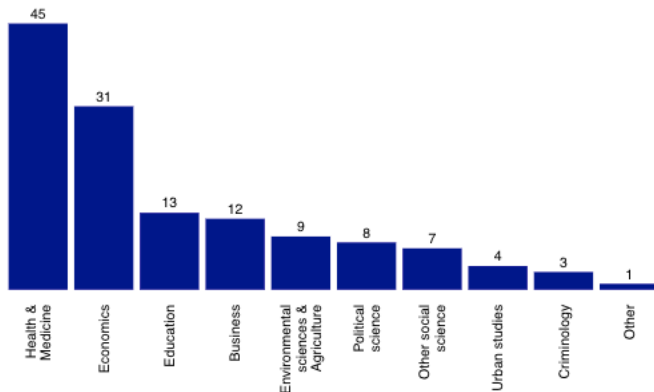
- 1 **Subsample:** Randomly draw a subsample $\mathcal{J} \subset \{1, \dots, n\}$ of size s without replacement.
- 2 **Grow Propensity Tree:**
 - ▶ Train a **classification tree** using (X_i, W_i) pairs in $\mathcal{J}$,
 - ▶ Splits are chosen to predict W_i (e.g., using CART's Gini criterion),
 - ▶ Each leaf must contain at least k units from *each treatment group*.
- 3 **Estimate Treatment Effect:**
 - ▶ For each test point x , estimate $\hat{\tau}(x)$ in its leaf using Equation (5).

Notes:

- Only treatment assignment W_i guides splits, not outcomes Y_i ,
- This avoids overfitting to outcomes and focuses on treatment heterogeneity.

A Survey of Recent Applications

Rehill(2024): *Causal forests are powerful for heterogeneous effect estimation but require critical attention to identification assumptions, transparent reporting of hyperparameters, and cautious adoption of innovative visualization methods. The field should move beyond “default practice” to leverage the methods full potential while ensuring statistical rigor.*



Implementing Causal Forest in ‘R’ and ‘Python’

R: grf package, tutorial-paper, [grf-doc](#)

Python: econml package, [econml-doc](#)

- A simulation study comparing the causal forest approach with mean-value based treatment effects
- Application to a real-world dataset: *The effect of US-China trade war on US firms' innovation*

References



A. Athey and G. W. Imbens. *Recursive partitioning for heterogeneous causal effects*. PNAS, 2016.



S. Wager and S. Athey. *Estimation and Inference of Heterogeneous Treatment Effects using Random Forests*. JASA, 2018.



Rehill, P., 2024. How do applied researchers use the Causal Forest? A methodological review of a method. <https://doi.org/10.48550/arXiv.2404.13356>

Appendix

Lipschitz Continuity

Lipschitz Continuous Function: A function $f(x)$ is **Lipschitz continuous** if there exists a constant $L < \infty$ such that for all x_1, x_2 in the domain of f :

$$|f(x_1) - f(x_2)| \leq L\|x_1 - x_2\|.$$

where $\|\cdot\|$ is a norm on the space of x , and L is the Lipschitz constant.

Appendix

Brief History of Jackknife

- **Origin:** Introduced by **Maurice Quenouille** in **1949**.
 - ▶ Original goal: **bias reduction** in estimators (e.g., ratio estimates).
- **Extended by John Tukey** in **1956**.
 - ▶ Emphasized its use for **variance estimation**.
 - ▶ Developed a general-purpose framework for resampling.
- **Name origin:** Coined by Tukey inspired by the *jackknife* pocket tool.
 - ▶ Symbolizes a **versatile, all-purpose tool** in statistics.
- **Legacy:**
 - ▶ Foundation for modern resampling methods like the *bootstrap* and *infinitesimal jackknife*.
 - ▶ Still widely used for variance estimation and sensitivity analysis.

